LA(1)LR(0)

임기정

Definition (LR(0)). Let $G \triangleq (N, T, P, S)$ be a CFG.

(i) $G' \triangleq (N', T', P', S') = (N \uplus \{S'\}, T \uplus \{\$\}, P \cup \{[S' \to S]\}, S')$.

(ii) $V \triangleq N \uplus T$. $V' \triangleq N' \uplus T'$.

(iii) $\alpha_1 \underset{\mathbf{rm}\ P'}{\Longrightarrow} \alpha_2 \overset{\text{def}}{\Longleftrightarrow} (\alpha_1, \alpha_2) \in \{(\alpha A z, \alpha\omega z) \mid [A \to \omega] \in P', \alpha \in (V')^*, z \in (T')^*\}$.

(iv) $L(G) \triangleq \left\{ z \in T^* \;\middle|\; S \Rightarrow^*_{\text{prev}} z \right\}$. $L(G') \triangleq \left\{ z \in T^* \;\middle|\; S' \underset{\mathbf{rm}\ P'}{\Longrightarrow}^+ z \right\}$. $L(G) = L(G')$.

(v) $I_{\text{LR}(0)} \triangleq \{[A \to \alpha \cdot \beta] \mid [A \to \alpha\beta] \in P'\}$.

(vi) $\text{Cl}(q) =_\mu q \cup \{[A \to \cdot\omega] \mid [A \to \omega] \in P', [B \to \alpha_1 \cdot A\alpha_2] \in \text{Cl}(q)\}$ for $q \in \mathcal{P}(I_{\text{LR}(0)})$.

(vii) $q_0 := \text{Cl}(\{[S' \to \cdot S]\})$.

(viii) $\text{GOTO}(q, X) := \text{Cl}(\{[A \to \alpha X \cdot \beta] \mid [A \to \alpha \cdot X\beta] \in q\})$ for $q \in \mathcal{P}(I_{\text{LR}(0)})$ and $X \in V$.

(ix) $Q \triangleq \text{Tbl} \setminus \{\emptyset\}$ where $\text{Tbl} =_\mu \{q_0\} \cup \{\text{GOTO}(q, X) \mid q \in \text{Tbl}, X \in V\}$.

(x) $\varepsilon : p \to q \overset{\text{def}}{\Longleftrightarrow} p \in Q \wedge p = q$. $X\alpha : p \to q \overset{\text{def}}{\Longleftrightarrow} p \in Q \wedge \alpha : \text{GOTO}(p, X) \to q$.

(xi) $\text{next}(q, X) := \text{GOTO}(q, X)$ for $q \in Q$ and $X \in V$.

(xii) $\text{reduce}(q, t) := \{[A \to \omega] \mid [A \to \omega\cdot] \in q, t \in \mathbf{LA}(q, [A \to \omega])\}$ for $q \in Q$ and $t \in T'$.

(xiii)
$$\frac{q'' = \text{next}(q', t) \quad q'' \neq \emptyset}{(\alpha : q \to q', tz) \vdash (\alpha t : q \to q'', z)}\ \text{Shift}$$

(xiv)
$$\frac{[A \to \omega] \in \text{reduce}(q', t) \quad q'' = \text{next}(q', A) \quad q'' \neq \emptyset}{(\alpha\omega : q \to q', tz) \vdash (\alpha A : q \to q'', tz)}\ \text{Reduce}(A \to \omega)$$

(xv) $q_f := \text{GOTO}(q_0, S)$.

(xvi) $L(LRA(G)) \triangleq \{z \in T^* \mid (\varepsilon : q_0 \to q_0, z\$) \vdash^+ (S : q_0 \to q_f, \$)\}$. $L(G) = L(LRA(G))$.

(xvii) For $ts_1 \subseteq \{z \in (T')^* \mid |z| \leq 1\}$ and $ts_2 \subseteq \{z \in (T')^* \mid |z| \leq 1\}$,
$$ts_1 \odot ts_2 := \begin{cases} (ts_1 \setminus \{\varepsilon\}) \cup ts_2, & \text{if } \varepsilon \in ts_1; \\ ts_1, & \text{if } \varepsilon \notin ts_1. \end{cases}$$

(xviii) $\text{First}(A) := \{t \mid t \in T', \omega \in V^*, A \Rightarrow^*_{P'} t\omega\} \cup \{\varepsilon \mid A \Rightarrow^*_{P'} \varepsilon\}$ for $A \in N'$.

(xix) $\text{First}(t) := \{t\}$ for $t \in T'$.

(xx) $\text{First}(\varepsilon) := \{\varepsilon\}$. $\text{First}(X\alpha) := \text{First}(X) \odot \text{First}(\alpha)$ for $X \in V'$ and $\alpha \in (V')^*$.

Theorem (LA(1)). $[A \to \omega\cdot] \in q$인 $q \in Q$, $A \in N'$와 $\omega \in V^*$에 대하여,

$$\mathbf{LA}(q, [A \to \omega]) \triangleq \left\{ t \in T' \;\middle|\; S' \underset{\mathbf{rm}\ P'}{\Longrightarrow}^* \alpha A z, \alpha\omega : q_0 \to q, \alpha \in V^*, z \in T^*, t \in \mathrm{First}(z\$) \right\} \tag{1}$$

로 두자. 이제 $\mathbf{LA}' \subseteq \{((q, [A \to \alpha \cdot \beta]), z) \mid q \in Q, [A \to \alpha \cdot \beta] \in q, z \in (T')^*, |z| \leq 1\}$를

$$\mathbf{LA}' =_\mu \mathbf{Propagate}_{\mathbf{LA}'}$$

로 정의하자. 그러면, $q \in Q \wedge [A \to \omega\cdot] \in q$일 때마다

$$\mathbf{LA}(q, [A \to \omega]) = \{t \in T' \mid ((q, [A \to \omega\cdot]), t) \in \mathbf{LA}'\} \tag{2}$$

이 성립한다. 여기서 **Propagate**의 도입규칙은 모두 3가지로,

$$M \,!\, k := \{v \mid (k, v) \in M\}$$

라고 하였을 때, 다음과 같은 것들이다:

$$\frac{}{\$ \in \mathbf{Propagate}_{\mathbf{LA}'_{\mathrm{prev}}} \,!\, (q_0, [S' \to \cdot S])} \tag{3}$$

$$\frac{}{\$ \in \mathbf{Propagate}_{\mathbf{LA}'_{\mathrm{prev}}} \,!\, (q_f, [S' \to S\cdot])} \tag{4}$$

$$\frac{q \in Q \quad [A \to \alpha \cdot \beta] \in q \quad A \neq S' \quad \alpha : p \to q \quad [B \to \alpha_1 \cdot A\alpha_2] \in p}{\mathrm{First}(\alpha_2) \odot \mathbf{LA}'_{\mathrm{prev}} \,!\, (p, [B \to \alpha_1 \cdot A\alpha_2]) \subseteq \mathbf{Propagate}_{\mathbf{LA}'_{\mathrm{prev}}} \,!\, (q, [A \to \alpha \cdot \beta])} \tag{5}$$

Proof. $q \in Q \wedge [A \to \alpha \cdot \beta] \in q$일 때마다

$$\mathbf{LA}' \,!\, (q, [A \to \alpha \cdot \beta]) = \bigcup \left\{ \mathrm{First}(z\$) \;\middle|\; S' \underset{\mathbf{rm}\ P'}{\Longrightarrow}^* \omega A z \underset{\mathbf{rm}\ P'}{\Longrightarrow} \omega\alpha\beta z, \omega\alpha : q_0 \to q, z \in T^* \right\} \tag{6}$$

이기 때문이다. □

감사의 인사

- 이 글을 쓰는 데 많은 도움을 주신 잇창명[1]님께 감사의 말씀을 드립니다.
- 또한, 참고문헌 [1]과 [2]로부터 따온 정의 및 정리를 사용하였음을 밝힙니다.

참고문헌

[1] Frank DeRemer and Thomas Pennello. "Efficient Computation of LALR(1) Look-Ahead Sets". In: *ACM Trans. Program. Lang. Syst.* 4.4 (Oct. 1982), pp. 615–649. ISSN: 0164-0925. DOI: `10.1145/69622.357187`. URL: `https://doi.org/10.1145/69622.357187`.

[2] 오세만. *컴파일러입문 (개정판)*. 정익사, 2010. ISBN: 978-8-93-530467-7.

[1] `https://eatchangmyeong.github.io/`